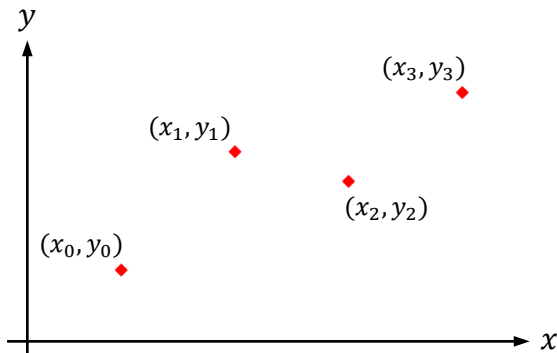


Approximation of Derivatives by Finite Difference



Definition of Derivative

$$f'(x) = \lim_{\Delta x \rightarrow 0} \frac{f(x + \Delta x) - f(x)}{\Delta x}$$

$$\text{or} \quad f'(x_0) = \lim_{x \rightarrow x_0} \frac{f(x) - f(x_0)}{x - x_0}$$

Finite Difference

$$f'(x) \approx \frac{f(x + \Delta x) - f(x)}{\Delta x}$$

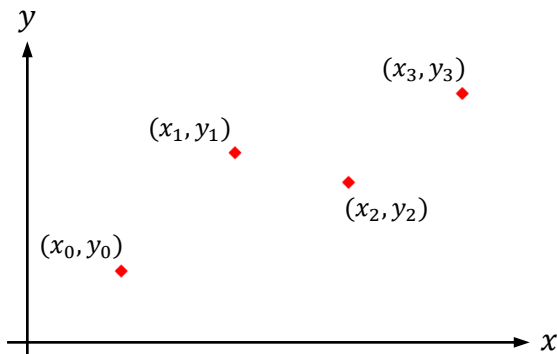
$$f'(x) \approx \frac{f(x - \Delta x) - f(x)}{-\Delta x}$$

$$f''(x) \approx \frac{f'(x + \Delta x) - f'(x)}{\Delta x}$$

$$\vdots$$

Forward Difference

Approximation of Derivatives by Finite Difference



Definition of Derivative

$$f'(x) = \lim_{\Delta x \rightarrow 0} \frac{f(x + \Delta x) - f(x)}{\Delta x}$$

or $f'(x_0) = \lim_{x \rightarrow x_0} \frac{f(x) - f(x_0)}{x - x_0}$

Finite Difference

$$f'(x) \approx \frac{f(x + \Delta x) - f(x)}{\Delta x}$$

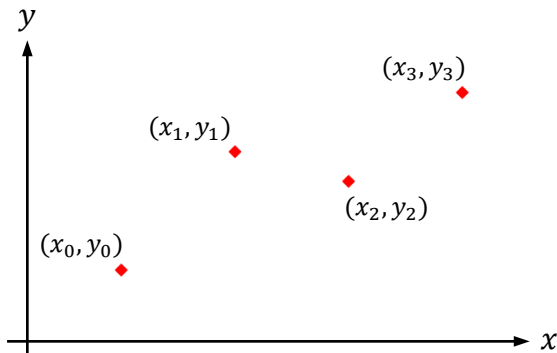
$$f'(x) \approx \frac{f(x) - f(x - \Delta x)}{\Delta x}$$

$$f''(x) \approx \frac{f'(x + \Delta x) - f'(x)}{\Delta x}$$

⋮

Forward Difference

Approximation of Derivatives by Finite Difference



Definition of Derivative

$$f'(x) = \lim_{\Delta x \rightarrow 0} \frac{f(x + \Delta x) - f(x)}{\Delta x}$$

$$\text{or} \quad f'(x_0) = \lim_{x \rightarrow x_0} \frac{f(x) - f(x_0)}{x - x_0}$$

Finite Difference

$$f'(x) \approx \frac{f(x + \Delta x) - f(x)}{\Delta x}$$

$$f'(x) \approx \frac{f(x) - f(x - \Delta x)}{\Delta x}$$

$$f''(x) \approx \frac{f'(x + \Delta x) - f'(x)}{\Delta x}$$

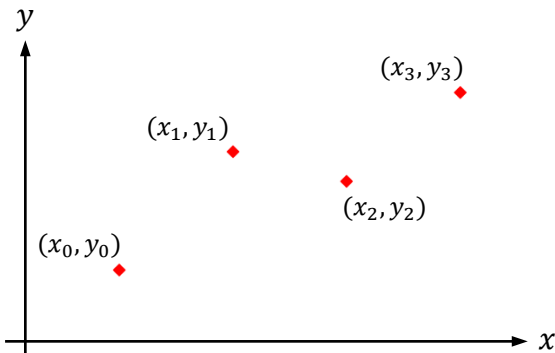
$$f''(x) \approx \frac{f'(x) - f'(x - \Delta x)}{\Delta x}$$

$$\vdots$$

$$\vdots$$

Forward Difference

Approximation of Derivatives by Finite Difference



Definition of Derivative

$$f'(x) = \lim_{\Delta x \rightarrow 0} \frac{f(x + \Delta x) - f(x)}{\Delta x}$$

or $f'(x_0) = \lim_{x \rightarrow x_0} \frac{f(x) - f(x_0)}{x - x_0}$

Finite Difference

$$f'(x) \approx \frac{f(x + \Delta x) - f(x)}{\Delta x}$$

$$f''(x) \approx \frac{f'(x + \Delta x) - f'(x)}{\Delta x}$$

$\vdots$

Forward Difference

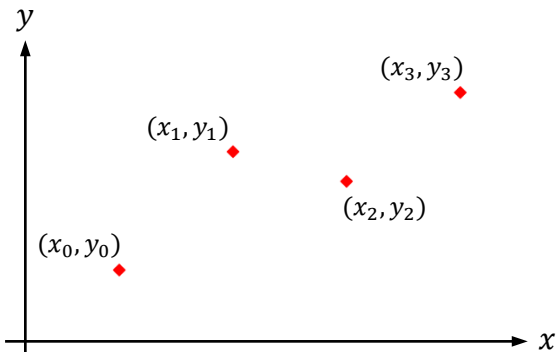
$$f'(x) \approx \frac{f(x) - f(x - \Delta x)}{\Delta x}$$

$$f''(x) \approx \frac{f'(x) - f'(x - \Delta x)}{\Delta x}$$

$\vdots$

Backward Difference

Approximation of Derivatives by Finite Difference



Definition of Derivative

$$f'(x) = \lim_{\Delta x \rightarrow 0} \frac{f(x + \Delta x) - f(x)}{\Delta x}$$

or
$$f'(x_0) = \lim_{x \rightarrow x_0} \frac{f(x) - f(x_0)}{x - x_0}$$

Finite Difference

$$f'(x) \approx \frac{f(x + \Delta x) - f(x)}{\Delta x}$$

$$f''(x) \approx \frac{f'(x + \Delta x) - f'(x)}{\Delta x}$$

⋮

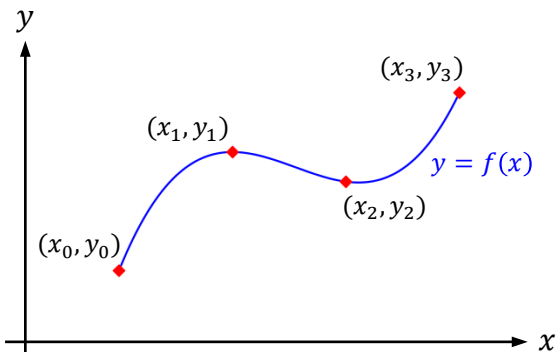
Forward Difference

$$f'(x) \approx \frac{f(x) - f(x - \Delta x)}{\Delta x}$$

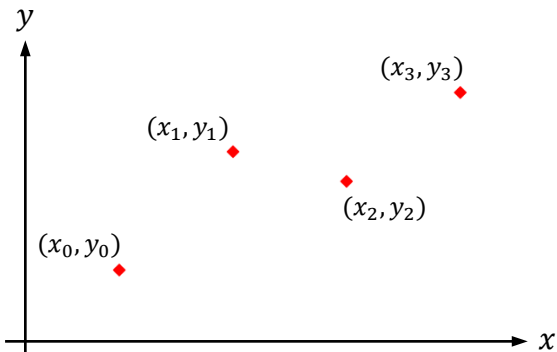
$$f''(x) \approx \frac{f'(x) - f'(x - \Delta x)}{\Delta x}$$

⋮

Backward Difference



Approximation of Derivatives by Finite Difference



Definition of Derivative

$$f'(x) = \lim_{\Delta x \rightarrow 0} \frac{f(x + \Delta x) - f(x)}{\Delta x}$$

or
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Finite Difference

$$f'(x) \approx \frac{f(x + \Delta x) - f(x)}{\Delta x}$$

$$f''(x) \approx \frac{f'(x + \Delta x) - f'(x)}{\Delta x}$$

⋮

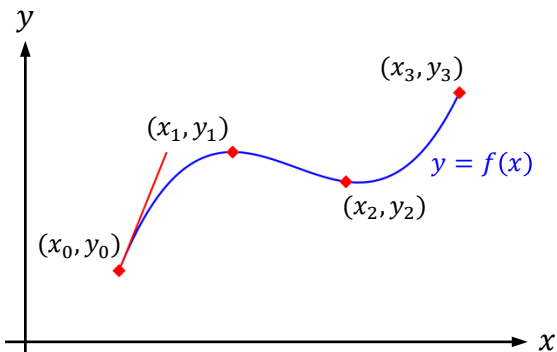
Forward Difference

$$f'(x) \approx \frac{f(x) - f(x - \Delta x)}{\Delta x}$$

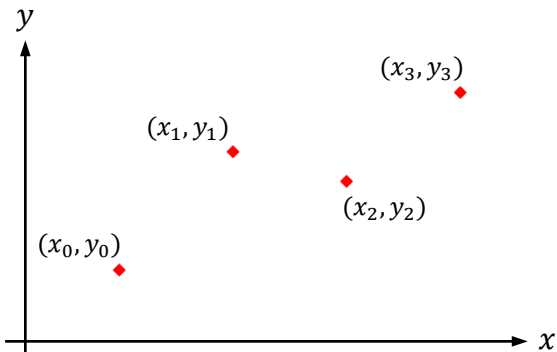
$$f''(x) \approx \frac{f'(x) - f'(x - \Delta x)}{\Delta x}$$

⋮

Backward Difference



Approximation of Derivatives by Finite Difference



Definition of Derivative

$$f'(x) = \lim_{\Delta x \rightarrow 0} \frac{f(x + \Delta x) - f(x)}{\Delta x}$$

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Finite Difference

$$f'(x) \approx \frac{f(x + \Delta x) - f(x)}{\Delta x}$$

$$f''(x) \approx \frac{f'(x + \Delta x) - f'(x)}{\Delta x}$$

⋮

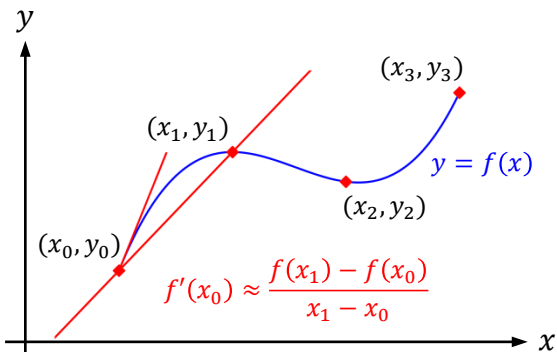
Forward Difference

$$f'(x) \approx \frac{f(x) - f(x - \Delta x)}{\Delta x}$$

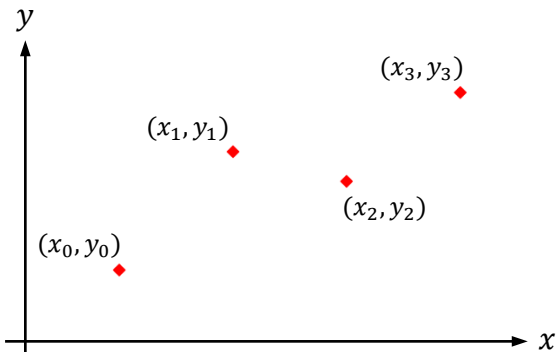
$$f''(x) \approx \frac{f'(x) - f'(x - \Delta x)}{\Delta x}$$

⋮

Backward Difference



Approximation of Derivatives by Finite Difference



Definition of Derivative

$$f'(x) = \lim_{\Delta x \rightarrow 0} \frac{f(x + \Delta x) - f(x)}{\Delta x}$$

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Finite Difference

$$f'(x) \approx \frac{f(x + \Delta x) - f(x)}{\Delta x}$$

$$f''(x) \approx \frac{f'(x + \Delta x) - f'(x)}{\Delta x}$$

⋮

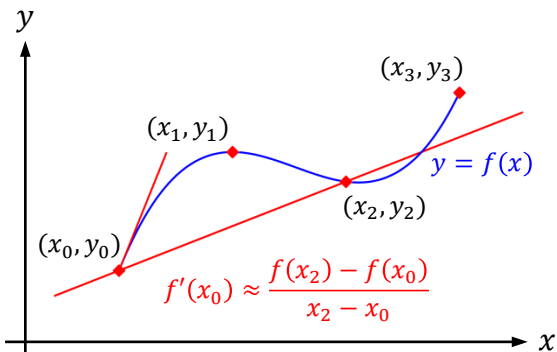
Forward Difference

$$f'(x) \approx \frac{f(x) - f(x - \Delta x)}{\Delta x}$$

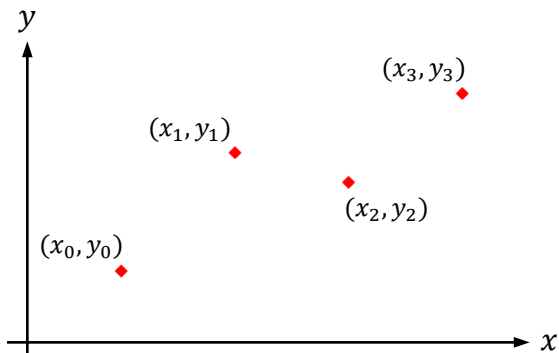
$$f''(x) \approx \frac{f'(x) - f'(x - \Delta x)}{\Delta x}$$

⋮

Backward Difference



Approximation of Derivatives by Finite Difference



Definition of Derivative

$$f'(x) = \lim_{\Delta x \rightarrow 0} \frac{f(x + \Delta x) - f(x)}{\Delta x}$$

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Finite Difference

$$f'(x) \approx \frac{f(x + \Delta x) - f(x)}{\Delta x}$$

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⋮

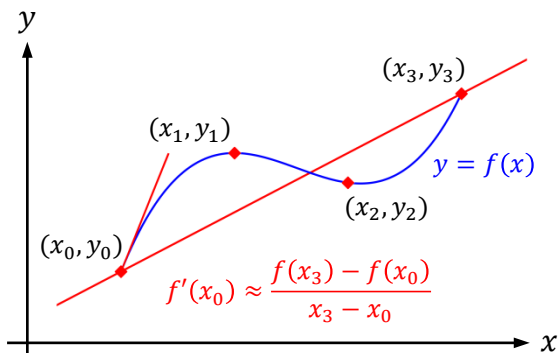
Forward Difference

$$f'(x) \approx \frac{f(x) - f(x - \Delta x)}{\Delta x}$$

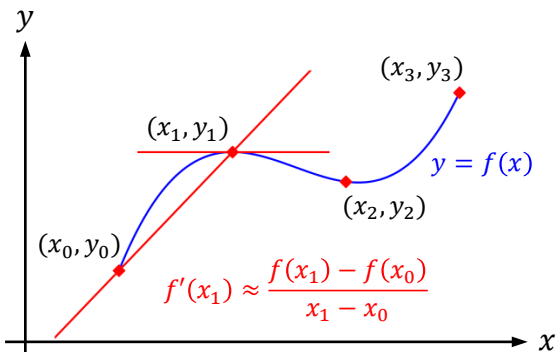
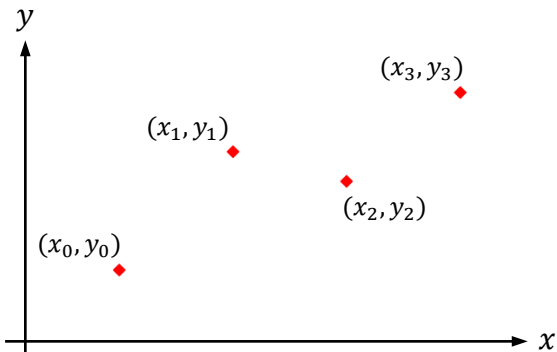
$$f''(x) \approx \frac{f'(x) - f'(x - \Delta x)}{\Delta x}$$

⋮

Backward Difference



Approximation of Derivatives by Finite Difference



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$$f'(x) = \lim_{\Delta x \rightarrow 0} \frac{f(x + \Delta x) - f(x)}{\Delta x}$$

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Finite Difference

$$f'(x) \approx \frac{f(x + \Delta x) - f(x)}{\Delta x}$$

$$f''(x) \approx \frac{f'(x + \Delta x) - f'(x)}{\Delta x}$$

⋮

Forward Difference

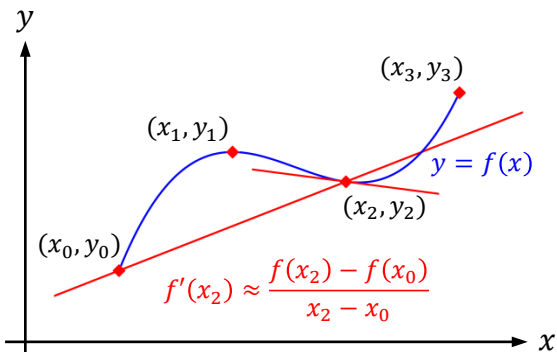
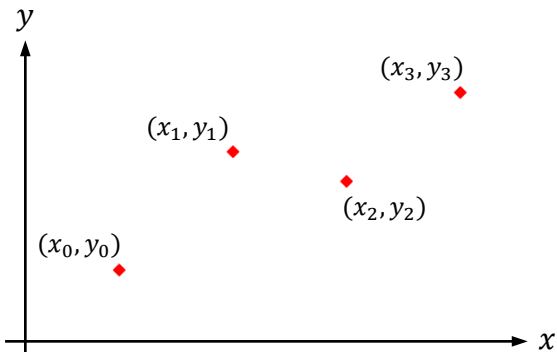
$$f'(x) \approx \frac{f(x) - f(x - \Delta x)}{\Delta x}$$

$$f''(x) \approx \frac{f'(x) - f'(x - \Delta x)}{\Delta x}$$

⋮

Backward Difference

Approximation of Derivatives by Finite Difference



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$$f'(x) = \lim_{\Delta x \rightarrow 0} \frac{f(x + \Delta x) - f(x)}{\Delta x}$$

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Finite Difference

$$f'(x) \approx \frac{f(x + \Delta x) - f(x)}{\Delta x}$$

$$f''(x) \approx \frac{f'(x + \Delta x) - f'(x)}{\Delta x}$$

⋮

Forward Difference

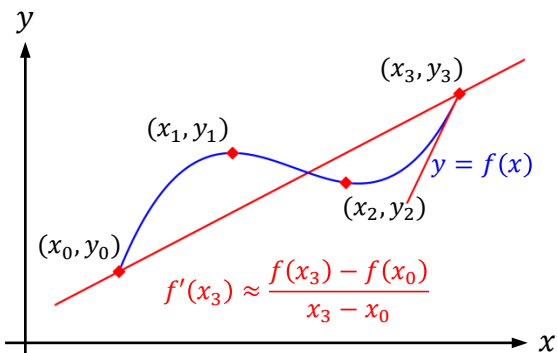
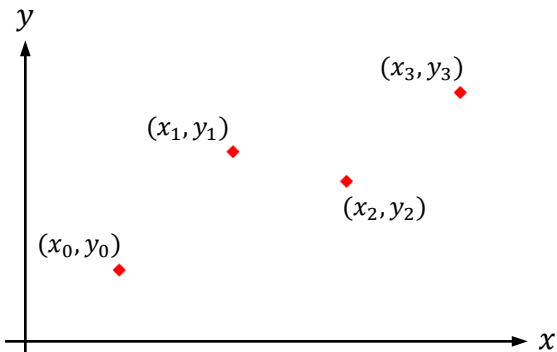
$$f'(x) \approx \frac{f(x) - f(x - \Delta x)}{\Delta x}$$

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⋮

Backward Difference

Approximation of Derivatives by Finite Difference



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$$f'(x) = \lim_{\Delta x \rightarrow 0} \frac{f(x + \Delta x) - f(x)}{\Delta x}$$

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Finite Difference

$$f'(x) \approx \frac{f(x + \Delta x) - f(x)}{\Delta x}$$

$$f''(x) \approx \frac{f'(x + \Delta x) - f'(x)}{\Delta x}$$

⋮

Forward Difference

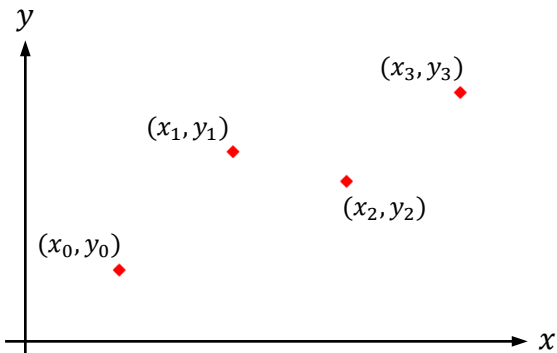
$$f'(x) \approx \frac{f(x) - f(x - \Delta x)}{\Delta x}$$

$$f''(x) \approx \frac{f'(x) - f'(x - \Delta x)}{\Delta x}$$

⋮

Backward Difference

Approximation of Derivatives by Finite Difference

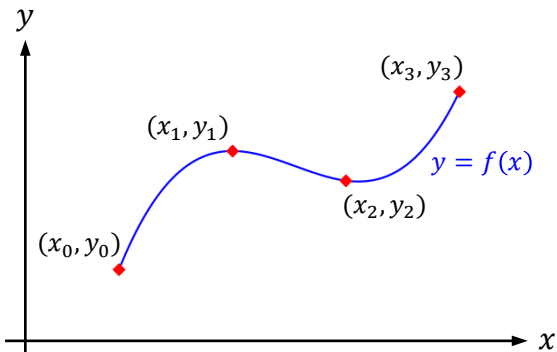


Definition of Derivative

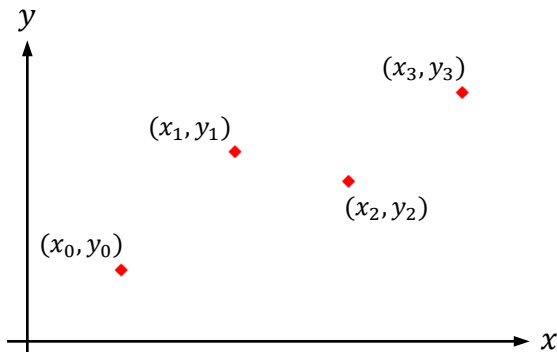
$$f'(x) = \lim_{\Delta x \rightarrow 0} \frac{f(x + \Delta x) - f(x)}{\Delta x}$$

$$\text{or } f'(x_0) = \lim_{x \rightarrow x_0} \frac{f(x) - f(x_0)}{x - x_0}$$

Finite Difference



Approximation of Derivatives by Finite Difference



Definition of Derivative

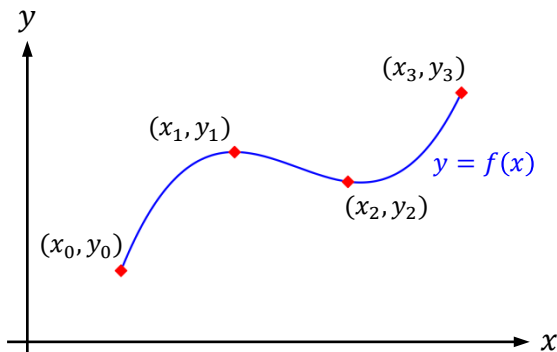
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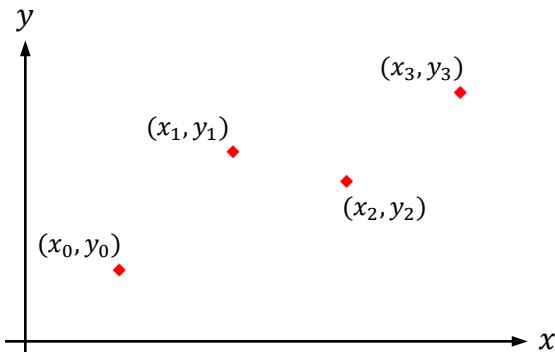
Finite Difference

$$f'(x) \approx \frac{f(x + \Delta x) - f(x)}{\Delta x}$$

Forward Difference



Approximation of Derivatives by Finite Difference



Definition of Derivative

$$f'(x) = \lim_{\Delta x \rightarrow 0} \frac{f(x + \Delta x) - f(x)}{\Delta x}$$

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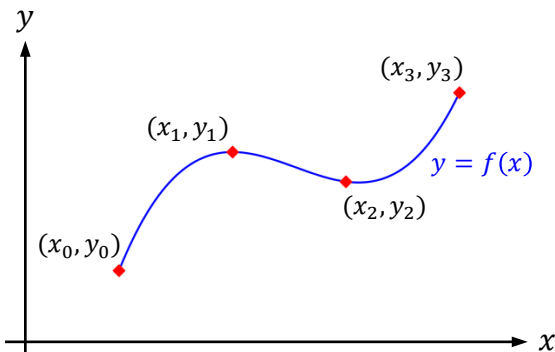
Finite Difference

$$f'(x) \approx \frac{f(x + \Delta x) - f(x)}{\Delta x}$$

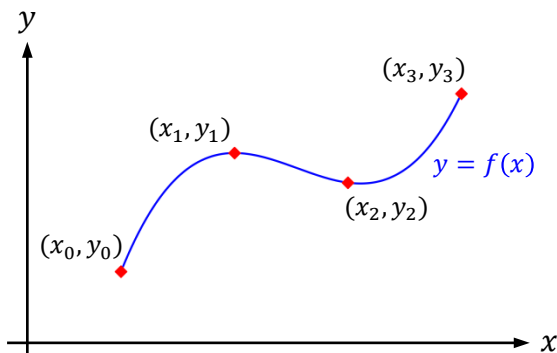
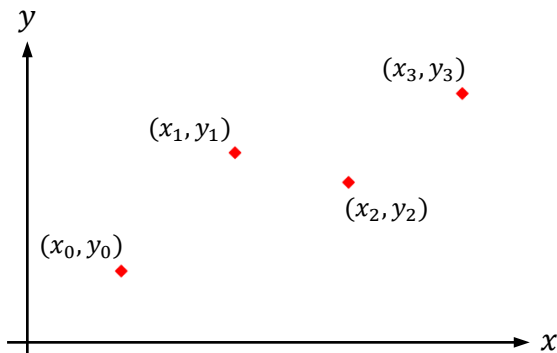
Forward Difference

$$f'(x) \approx \frac{f(x) - f(x - \Delta x)}{\Delta x}$$

Backward Difference



Approximation of Derivatives by Finite Difference



Definition of Derivative

$$f'(x) = \lim_{\Delta x \rightarrow 0} \frac{f(x + \Delta x) - f(x)}{\Delta x} \quad \text{or} \quad f'(x_0) = \lim_{x \rightarrow x_0} \frac{f(x) - f(x_0)}{x - x_0}$$

Finite Difference

$$f'(x) \approx \frac{f(x + \Delta x) - f(x)}{\Delta x}$$

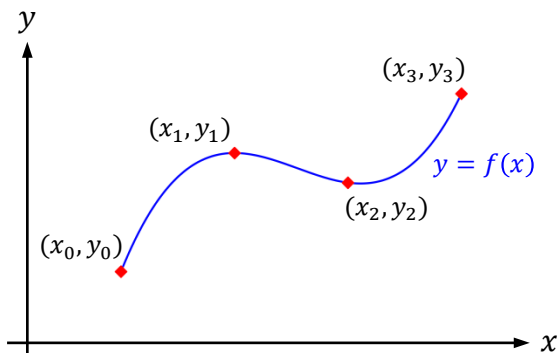
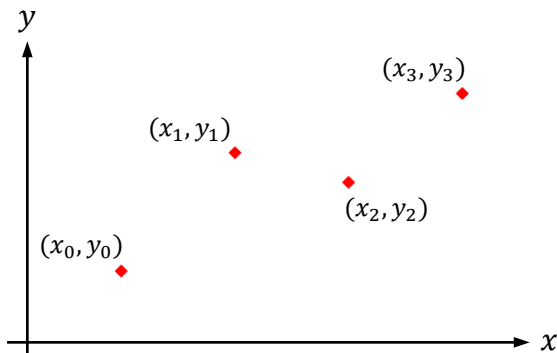
Forward Difference

$$f'(x) \approx \frac{f(x) - f(x - \Delta x)}{\Delta x}$$

Backward Difference

$$2 \cdot f'(x) \approx \frac{f(x + \Delta x) - f(x - \Delta x)}{\Delta x}$$

Approximation of Derivatives by Finite Difference



Definition of Derivative

$$f'(x) = \lim_{\Delta x \rightarrow 0} \frac{f(x + \Delta x) - f(x)}{\Delta x}$$

$$\text{or } f'(x_0) = \lim_{x \rightarrow x_0} \frac{f(x) - f(x_0)}{x - x_0}$$

Finite Difference

$$f'(x) \approx \frac{f(x + \Delta x) - f(x)}{\Delta x}$$

Forward Difference

$$f'(x) \approx \frac{f(x) - f(x - \Delta x)}{\Delta x}$$

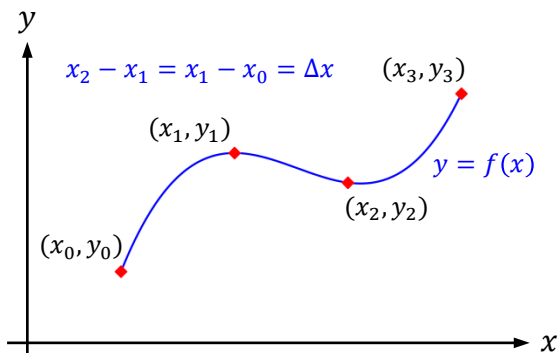
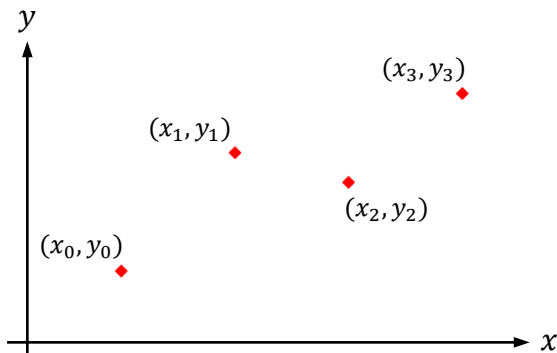
Backward Difference

$$2 \cdot f'(x) \approx \frac{f(x + \Delta x) - f(x - \Delta x)}{\Delta x}$$

$$f'(x) \approx \frac{f(x + \Delta x) - f(x - \Delta x)}{2 \cdot \Delta x}$$

Central Difference

Approximation of Derivatives by Finite Difference



Definition of Derivative

$$f'(x) = \lim_{\Delta x \rightarrow 0} \frac{f(x + \Delta x) - f(x)}{\Delta x}$$

$$\text{or } f'(x_0) = \lim_{x \rightarrow x_0} \frac{f(x) - f(x_0)}{x - x_0}$$

Finite Difference

$$f'(x) \approx \frac{f(x + \Delta x) - f(x)}{\Delta x}$$

Forward Difference

$$f'(x) \approx \frac{f(x) - f(x - \Delta x)}{\Delta x}$$

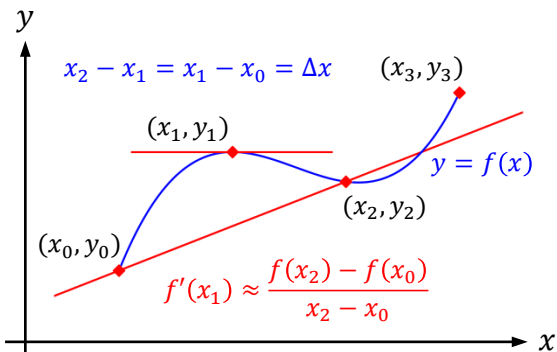
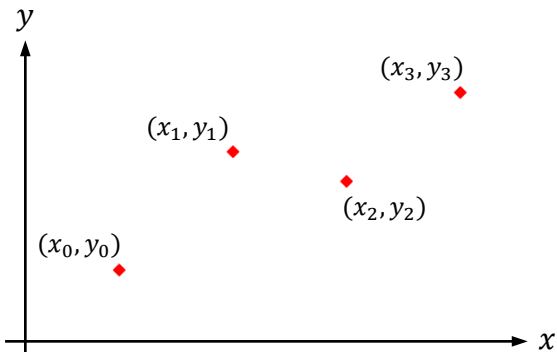
Backward Difference

$$2 \cdot f'(x) \approx \frac{f(x + \Delta x) - f(x - \Delta x)}{\Delta x}$$

$$f'(x) \approx \frac{f(x + \Delta x) - f(x - \Delta x)}{2 \cdot \Delta x}$$

Central Difference

Approximation of Derivatives by Finite Difference



Definition of Derivative

$$f'(x) = \lim_{\Delta x \rightarrow 0} \frac{f(x + \Delta x) - f(x)}{\Delta x}$$

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Finite Difference

$$f'(x) \approx \frac{f(x + \Delta x) - f(x)}{\Delta x}$$

Forward Difference

$$f'(x) \approx \frac{f(x) - f(x - \Delta x)}{\Delta x}$$

Backward Difference

$$2 \cdot f'(x) \approx \frac{f(x + \Delta x) - f(x - \Delta x)}{\Delta x}$$

$$f'(x) \approx \frac{f(x + \Delta x) - f(x - \Delta x)}{2 \cdot \Delta x}$$

Central Difference